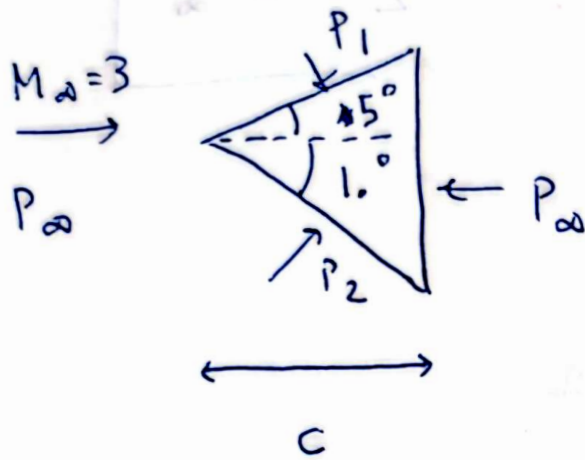
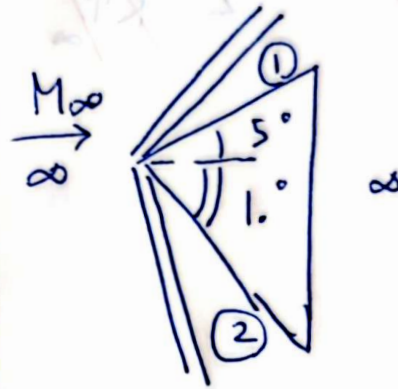


WED-06-11-2025, Sample question on Shock waves

Ex.1) Find c_p and c_d .



Solution:



upper part

$$M_{\infty} = 3 \quad \delta = 5^{\circ} \quad \xrightarrow[\text{calc}]{\text{online}} \quad \theta = 23.13 \rightarrow \frac{P_1}{P_{\infty}} = 1.45 \rightarrow \begin{cases} c_{p1} = \frac{P_1 - P_{\infty}}{q_{\infty}} \\ M_{n1} = 2.75 \\ c_{p1} = \frac{P_1 - P_{\infty}}{\frac{1}{2} \rho P_{\infty} M_{\infty}^2} \end{cases}$$

lower part

$$\begin{cases} M_{\infty} = 3 \\ \delta = 1^{\circ} \end{cases} \xrightarrow[\text{calc}]{\text{online}} \quad \theta = 27.38 \rightarrow \frac{P_2}{P_{\infty}} = 2.05 \quad \begin{cases} c_{p1} = \frac{2}{\gamma M_{\infty}^2} \left(\frac{P_1}{P_{\infty}} - 1 \right) = 0.071 \\ c_{p2} = 0.167 \end{cases}$$

Assume $P_{\infty} = 101.3 \text{ kPa} \rightarrow P_1 = 1.45 \times 101.3 \text{ k} = 146.88 \text{ kPa}$

$P_2 = 2.05 \times \text{ " } = 207.67 \text{ "}$

$$D = P_1 \sin \delta_1 \frac{c}{\cos \delta_1} + P_2 \sin \delta_2 \frac{c}{\cos \delta_2} - P_{\infty} c (\tan \delta_1 + \tan \delta_2)$$

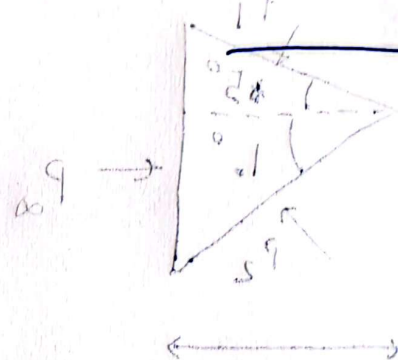
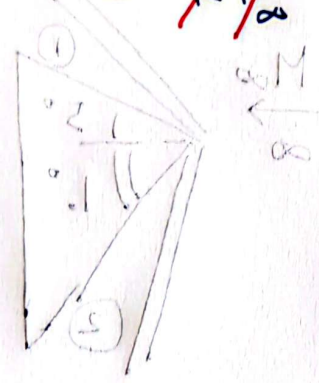
$$= 12.85c + 37.38c - 27.04c = 23.19c$$

$$c_d = \frac{D}{\frac{1}{2} \rho P_{\infty} M_{\infty}^2 c} = 0.036$$

(P-1)

Flow starts as uniform along x, 2500 ft/sec

$$q_\infty = \frac{1}{2} \gamma_\infty V_\infty^2 = \frac{1}{2} \frac{P_\infty}{\cancel{\gamma} \cancel{T}_\infty} M_\infty^2 \overbrace{\cancel{\gamma} \cancel{T}_\infty}^{a_\infty^2} = \frac{1}{2} \gamma P_\infty M_\infty^2$$



upper part

$$\frac{P_1}{P_\infty} = 1.47 = \frac{P_1}{P_\infty} \rightarrow P_1 = 1.47 \times 101.3 \text{ kPa} = 148.95 \text{ kPa}$$

$$\frac{P_2}{P_\infty} = 5.75 \rightarrow P_2 = 5.75 \times 101.3 \text{ kPa} = 582.47 \text{ kPa}$$

lower part

$$\frac{P_3}{P_\infty} = 5.75 \rightarrow P_3 = 5.75 \times 101.3 \text{ kPa} = 582.47 \text{ kPa}$$

$$\frac{P_4}{P_\infty} = 1.47 \rightarrow P_4 = 1.47 \times 101.3 \text{ kPa} = 148.95 \text{ kPa}$$

Pressure

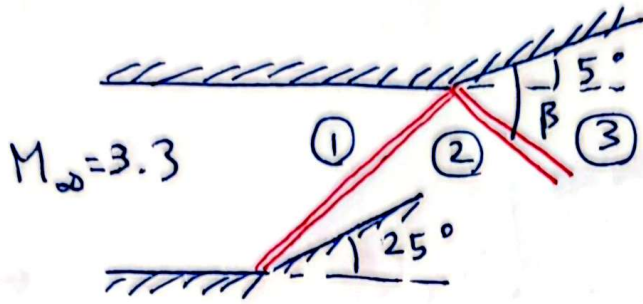
$$P_\infty = 101.3 \text{ kPa} \rightarrow P_1 = 1.47 \times 101.3 \text{ kPa} = 148.95 \text{ kPa}$$

$$P_2 = 5.75 \times 101.3 \text{ kPa} = 582.47 \text{ kPa}$$

Force

$$D = P_1 \cdot A_1 + P_2 \cdot A_2 + P_3 \cdot A_3 + P_4 \cdot A_4$$

Ex. 2



Find $\frac{T_3}{T_1}$, $\frac{P_3}{P_1}$, M_3 and β

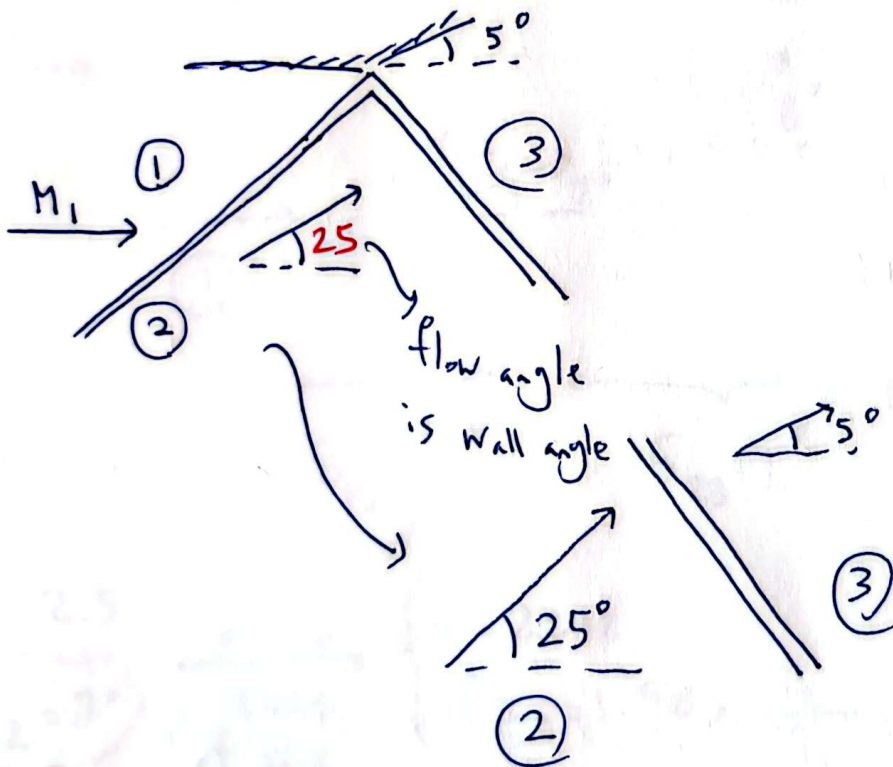
Solution:

$$M_1 = 3.3$$

$$\delta_1 = 25^\circ$$

$$\xrightarrow[\text{calc}]{\text{online}} \theta_1 = 41.88$$

$$\begin{cases} M_2 = 1.38 \\ \frac{P_2}{P_1} = 5.99 \\ \frac{T_2}{T_1} = 1.86 \end{cases}$$



$$\begin{cases} \delta_{2-3} = 25 - 5 = 20 \\ M_2 = 1.38 \end{cases}$$

$$\xrightarrow[\text{calc}]{\text{online}}$$

$$\begin{cases} \theta = 58.99^\circ, M_3 = 1.055 \\ \frac{P_3}{P_2} = 2.37, \frac{T_3}{T_2} = 1.4 \end{cases} = \beta$$

$$\frac{P_3}{P_1} = \frac{P_3}{P_2} \times \frac{P_2}{P_1} = 15.76$$

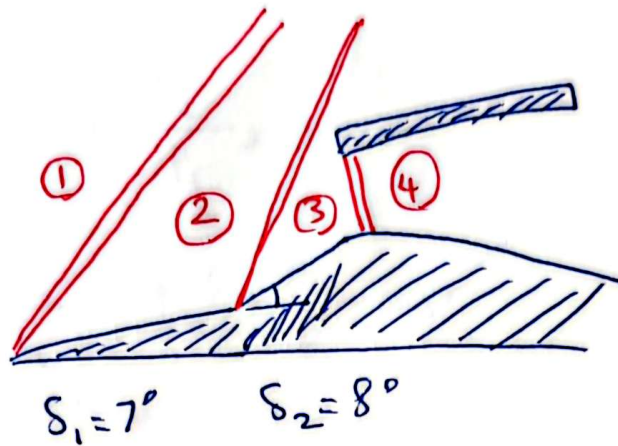
$$\frac{T_3}{T_1} = \frac{T_3}{T_2} \times \frac{T_2}{T_1} = 2.6$$

✱

(P-2)

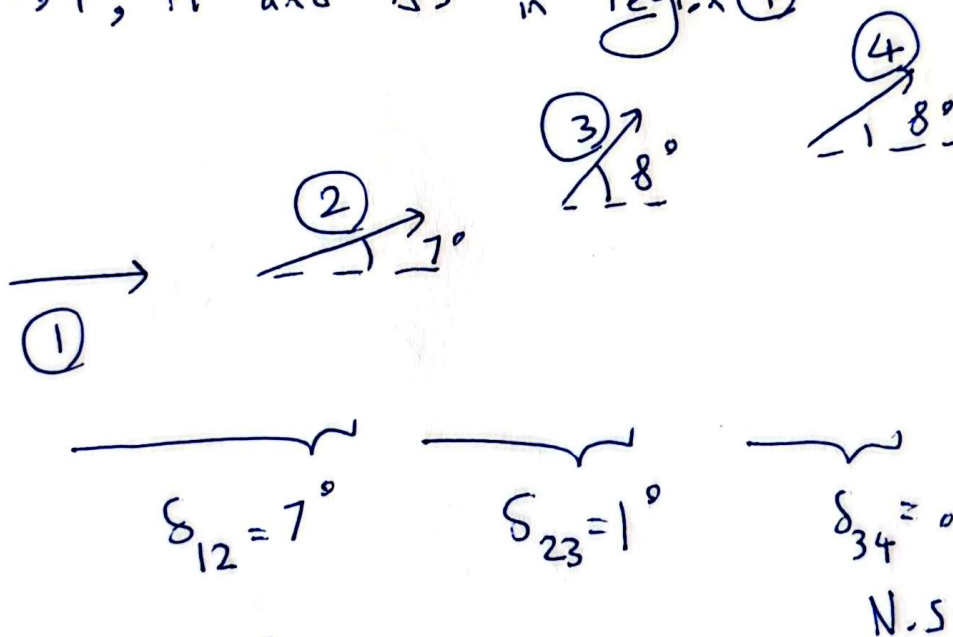
Ex. 3

$M_\infty = 2.5$
 $P_\infty = 243 \text{ psia}$
 $T_\infty = 392^\circ \text{R}$



Find T, P, M and Δs in region (4)

Solution:



$M_1 = 2.5$
 $\delta_{12} = 7^\circ$

online calc of 0.5 $\left\{ \begin{array}{l} \theta = 29.12 \\ \frac{P_2}{P_1} = 1.56, \frac{T_2}{T_1} = 1.14, M_2 = 2.21 \end{array} \right.$

$M_2 = 2.21$
 $\delta_{23} = 1^\circ$

online calc of 0.5 $\left\{ \begin{array}{l} \theta = \checkmark \\ \frac{P_3}{P_2} = \checkmark \end{array} \right.$

$\frac{T_3}{T_2} = \checkmark$

$M_3 = \checkmark$

$M_3 = \checkmark$
 N.S

online calc of N.S $\rightarrow \left\{ \begin{array}{l} \frac{P_4}{P_3} = \checkmark \\ \frac{T_4}{T_3} = \checkmark \end{array} \right.$

$M_4 = \checkmark$

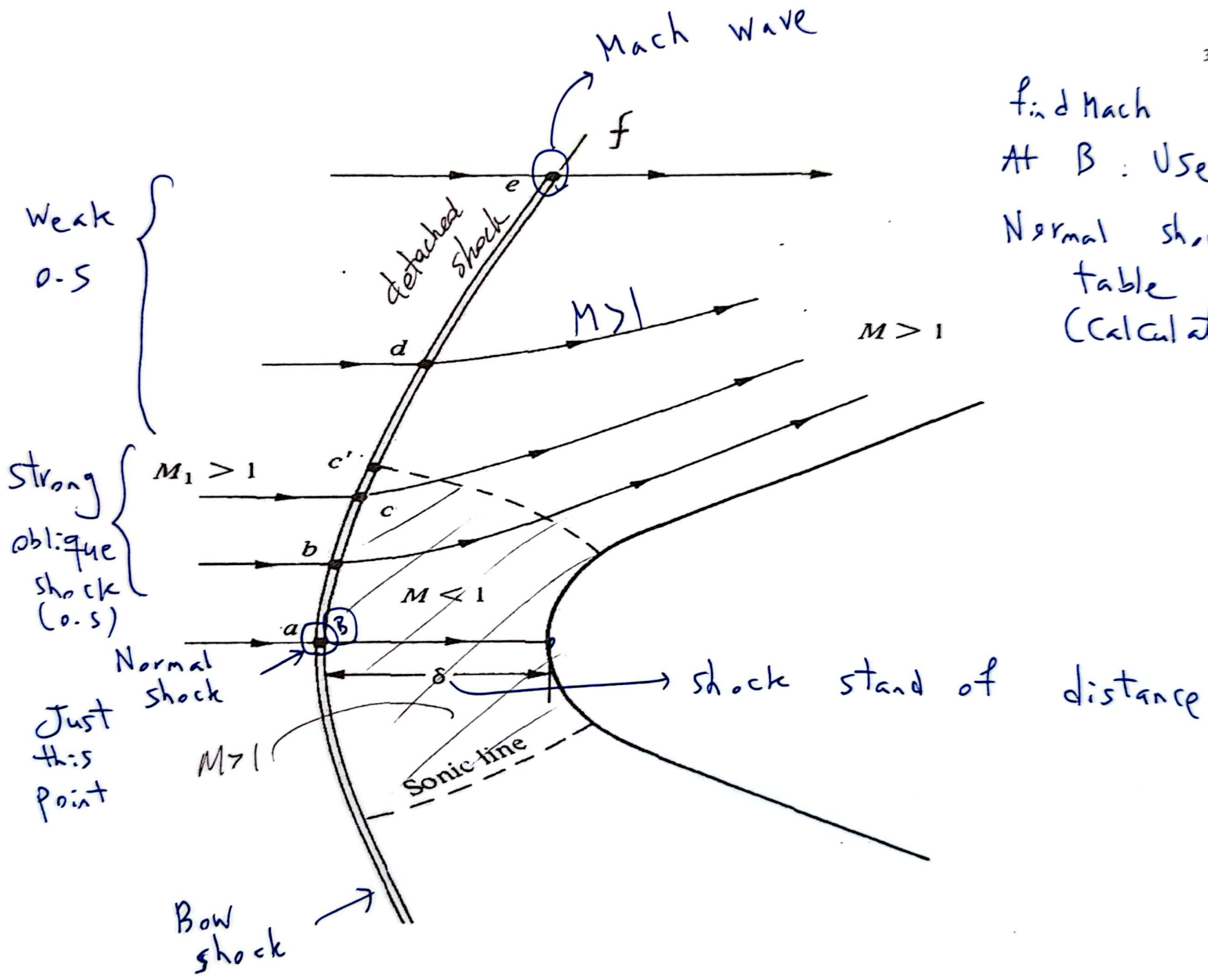
(P-3)

$$T_4 = \frac{T_4}{T_3} \times \frac{T_3}{T_2} \times \frac{T_2}{T_1} \times T_1 = \checkmark$$

$$P_4 = \frac{P_4}{P_3} \times \frac{P_3}{P_2} \times \frac{P_2}{P_1} \times P_1 = \checkmark$$

$$\Delta S = \frac{\gamma R}{\delta - 1} \ln \frac{T_4}{T_1} - R \ln \frac{P_4}{P_1} = \checkmark$$

find Mach
At B : Use
Normal shock
table
(calculator)



Detached shock wave in front of blunt body

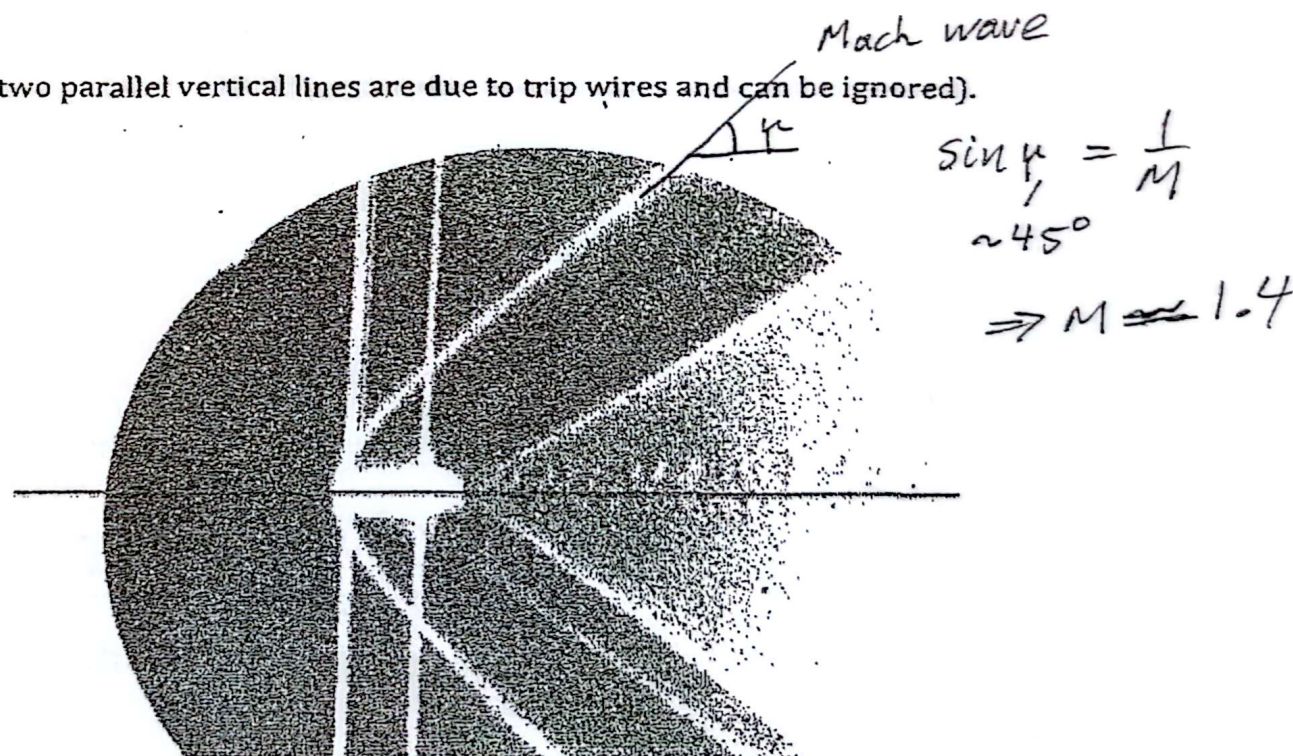
- a. normal
- b. oblique shock (strong)
- c. dividing point between strong-weak solution
- d. flow supersonic behind the shock
- e. weak solution
- f. Mach wave (*far-field solution*)

shape of shock and detachment distance δ can be found only from numerical solutions.

The first image of supersonic flow was taken by Ernst Mach and is shown below. It has been first published in 1887 and is often reproduced in textbooks without specifying the free-stream Mach number (since the concept was not introduced until 1920s).

Based on the information in the figure, estimate the free-stream Mach number for this flow.

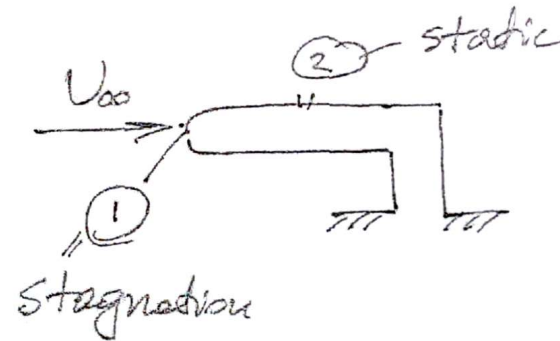
(Note: two parallel vertical lines are due to trip wires and can be ignored).



Velocity Measurement

(Pitot-static tube)

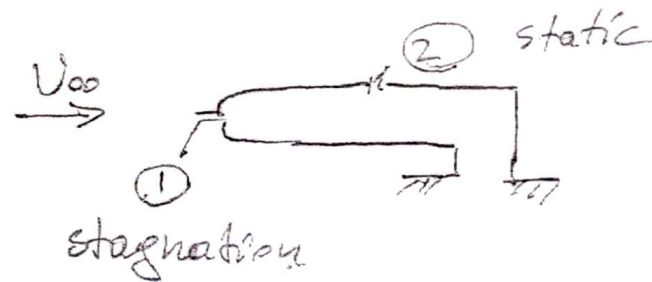
a. Incompressible (201)



Bernoulli's equation: $p_1 + \frac{1}{2} \cancel{\rho u_1^2} = p_2 + \frac{1}{2} \rho u_2^2 \rightarrow u_2 = U_\infty$
 $u_1 = 0$

$$\Rightarrow U_\infty = \sqrt{\frac{2}{\rho}(p_1 - p_2)} \quad (\text{need to know } \rho)$$

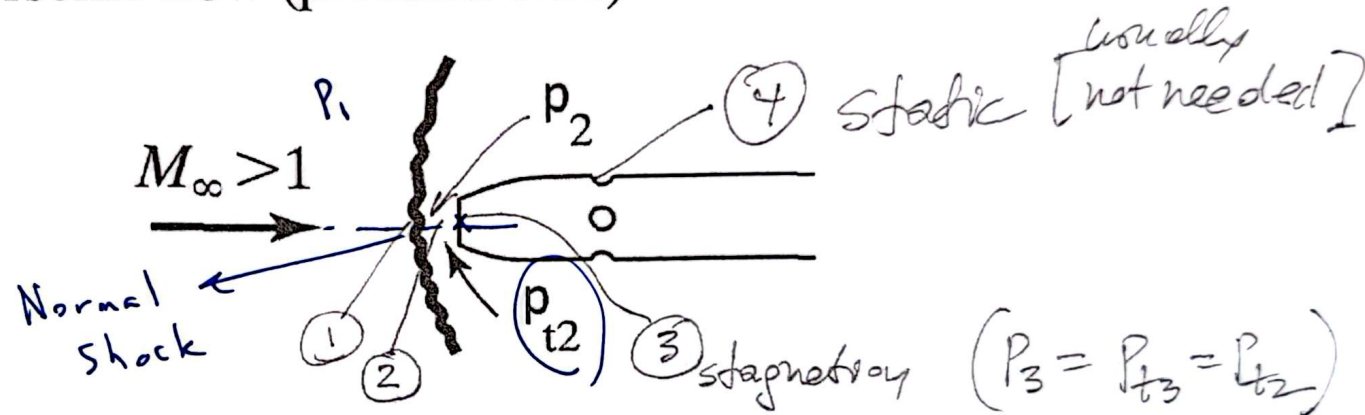
b. Compressible subsonic



Assume isentropic flow: ^(8.31) $\frac{p_t = p_1}{p = p_2} = \left(1 + \frac{\gamma - 1}{2} M_\infty^2 \right)^{\gamma/(\gamma - 1)}$

$$\Rightarrow M_\infty^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_1}{p_2} \right)^{(\gamma - 1)/\gamma} - 1 \right]$$

c. b. Supersonic flow (problem 8.11)



Isentropic $\frac{p_{t2}}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_2^2\right)^{\gamma(\gamma-1)}$ (8.36)

Across shock $\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1)$ (8.75)
normal

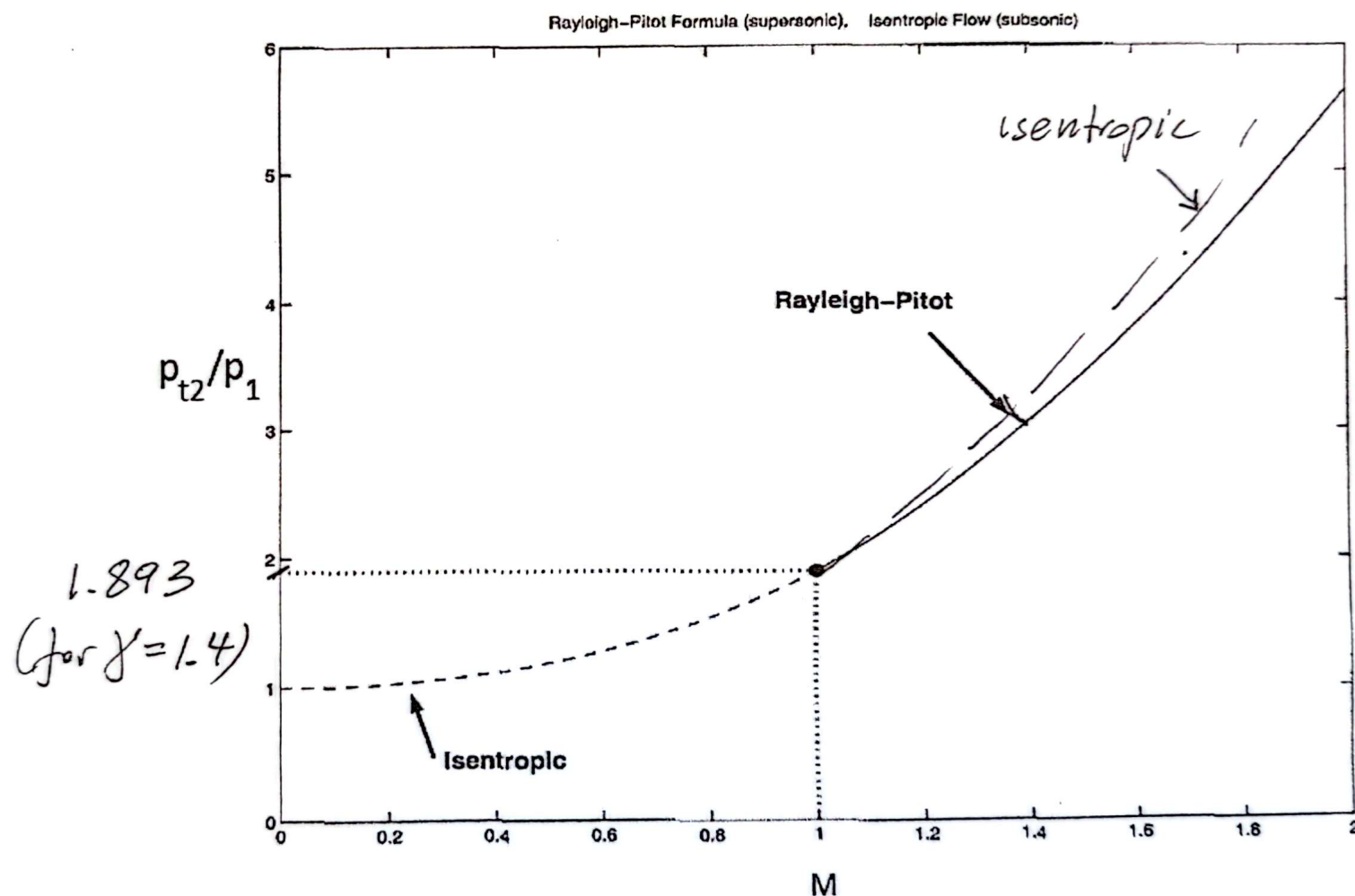
$M_2^2 = \frac{1 + [(\gamma - 1)/2] M_1^2}{\gamma M_1^2 - (\gamma - 1)/2}$ $\rightarrow$ M after the shock

$$\Rightarrow \frac{p_{t2}}{p_1} = \frac{p_{t2}}{p_2} \frac{p_2}{p_1}$$

Substitute: ✓

$$\Rightarrow \frac{p_{t2}}{p_1} = \left(\frac{(\gamma + 1)^2 M_1^2}{4\gamma M_1^2 - 2(\gamma - 1)} \right)^{\gamma/(\gamma-1)} \frac{1 - \gamma + 2\gamma M_1^2}{\gamma + 1} = f(M_1)$$

← Rayleigh-Pitot tube formula - can be part of shock tables



given $\frac{p_{t2}}{p_1}$ check if $>$ or $<$ than 1.893

$\nwarrow$ $\nearrow$

R-P formula ($M > 1$) isentropic ($M < 1$)

Example: A Pitot tube is placed in an air stream. The freestream pressure is $P_\infty = 1 \text{ atm} = 101,325 \text{ Pa}$. If the Pitot tube measures a value of stagnation pressure of $1,267,000 \text{ Pa}$, determine the free stream Mach number, M_∞ .

$$\frac{P_{t2}}{P_1} = \frac{1,267,000}{101,325} = 12.5043 \quad (> 1.893)$$

$$P-R \text{ formula: } \leadsto M_1 = 3.05$$

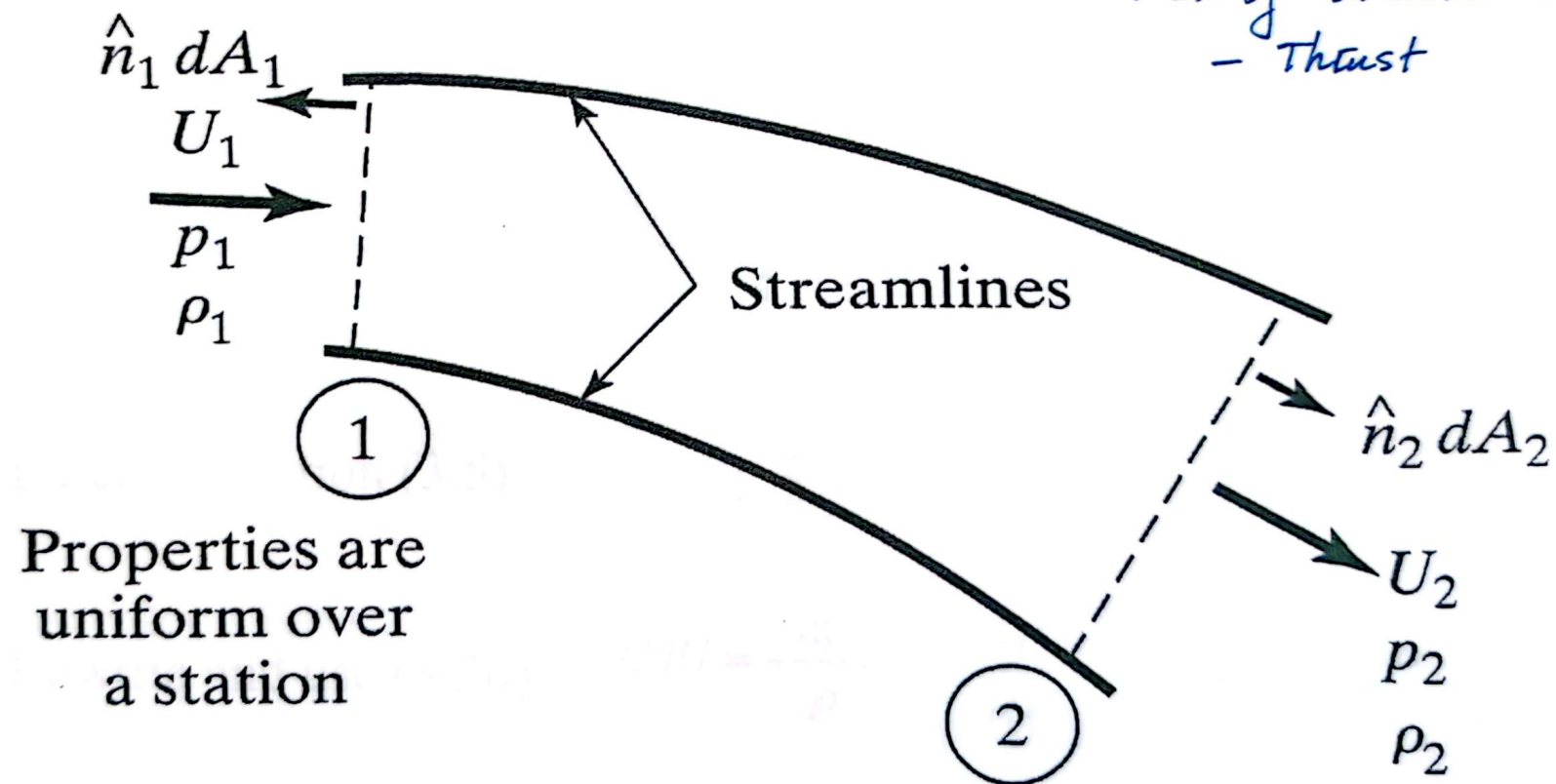
$$\left[\text{note: for isentropic flow} = \frac{P_{01}}{P_1} = 12.5 \leadsto M = 2.3 \right]$$

wrong answer!

Compressible Flow Through Nozzles [8.3, 8.4] + notes

Objective: Study flows in converging diverging nozzles

*including shock waves
- Thrust*



Conservation of mass: $\rho_1 U_1 A_1 = \rho_2 U_2 A_2$

$$\rho U A = \text{constant} \Rightarrow d(\rho U A) = 0$$

$$(d\rho)UA + \rho(dU)A + \rho U(dA) = 0$$

$$\Rightarrow \frac{d\rho}{\rho} + \frac{dU}{U} + \frac{dA}{A} = 0 \quad (8.49)$$

Euler's equation (3.5) $d\left(\frac{U^2}{2}\right) + \frac{dp}{\rho} = 0$

Rewrite and use (8.26) $UdU = -\frac{dp}{\rho} = -\frac{d\rho}{d\rho} \cdot \frac{d\rho}{\rho} = -a^2 \frac{d\rho}{\rho}$

$$\Rightarrow \frac{dp}{\rho} = -\frac{UdU}{a^2} = -M^2 \frac{dU}{U}$$

$$\left(-M^2 \frac{dU}{U} + \frac{dU}{U} + \frac{dA}{A} = 0 \Rightarrow \right)$$

put back to (8.49) $(1-M^2) \frac{dU}{U} + \frac{dA}{A} = 0$

$$\boxed{\frac{dU}{U} = -\frac{dA/A}{(1-M^2)}}$$

Variation of velocity
velocity

$$= \frac{-\text{Variation of Area}}{\text{Area} (1-M^2)} \quad (8.50)$$

1) $0 \leq M < 1$ (subsonic flow) $\Rightarrow 1-M^2 > 0$

for $dU > 0 \Rightarrow dA < 0$

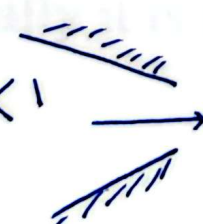
for $dU < 0 \Rightarrow dA > 0$

 converging

 diverging

To increase velocity you need a convergent duct

$M = \dots$ $\rho U A = \text{cte}$
 $U \propto \frac{1}{A}$

$M < 1$  M : increases in convergent nozzle
 M : decreases in divergent "

same trends as $\rho = \text{ct}$ flow

Note: for $M = 0 \Rightarrow \frac{dA}{A} = -\frac{dU}{U} \Rightarrow AU = \text{constant}$

2) For $M > 1$ (supersonic flow)

for $dU > 0 \Rightarrow dA > 0$



for $dU < 0 \Rightarrow dA < 0$

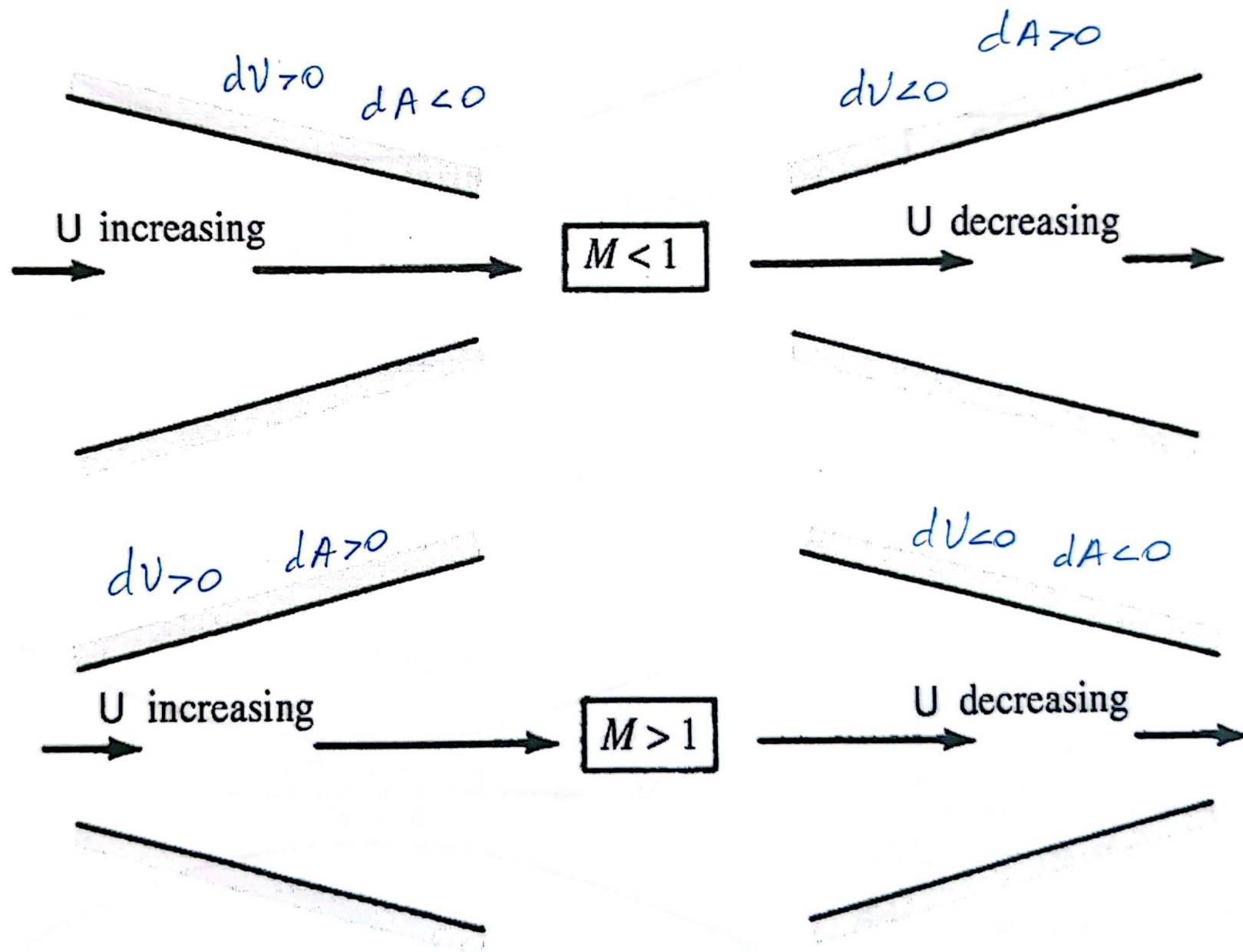


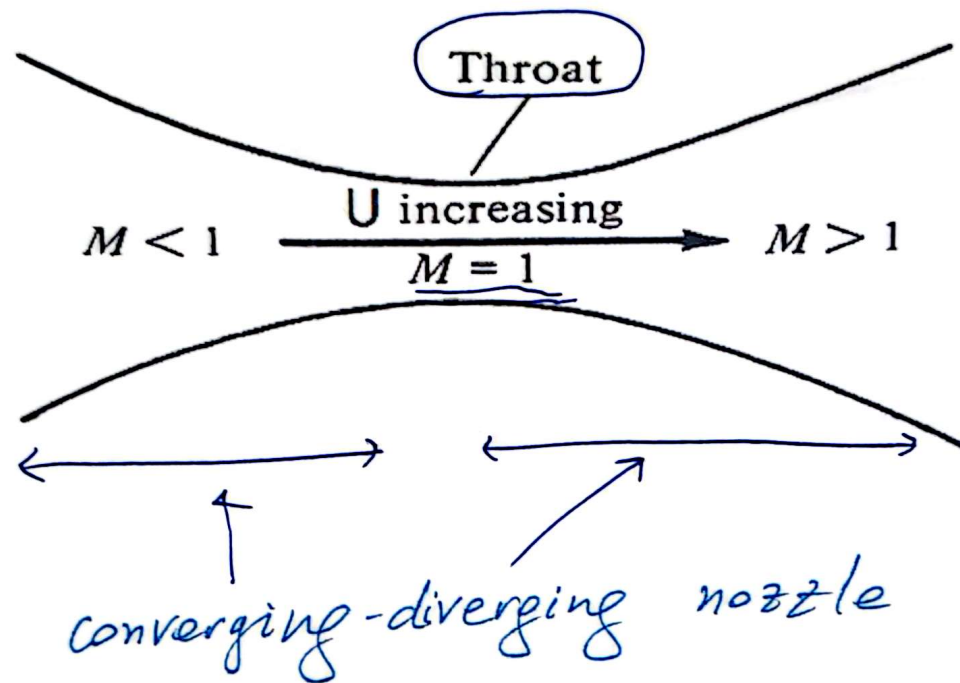
To increase velocity you need a divergent duct (opposite to $M < 1$)

3) $M = 1$ (sonic flow)

$$dA = 0 \quad \text{for} \quad dU \neq 0 \quad \Rightarrow \quad \text{max or min in } A(x)$$

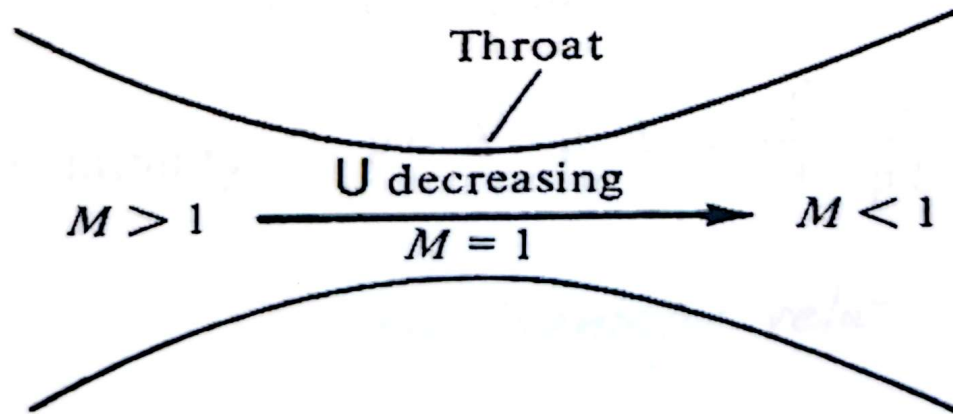
Actually it is a min, convergent-divergent duct





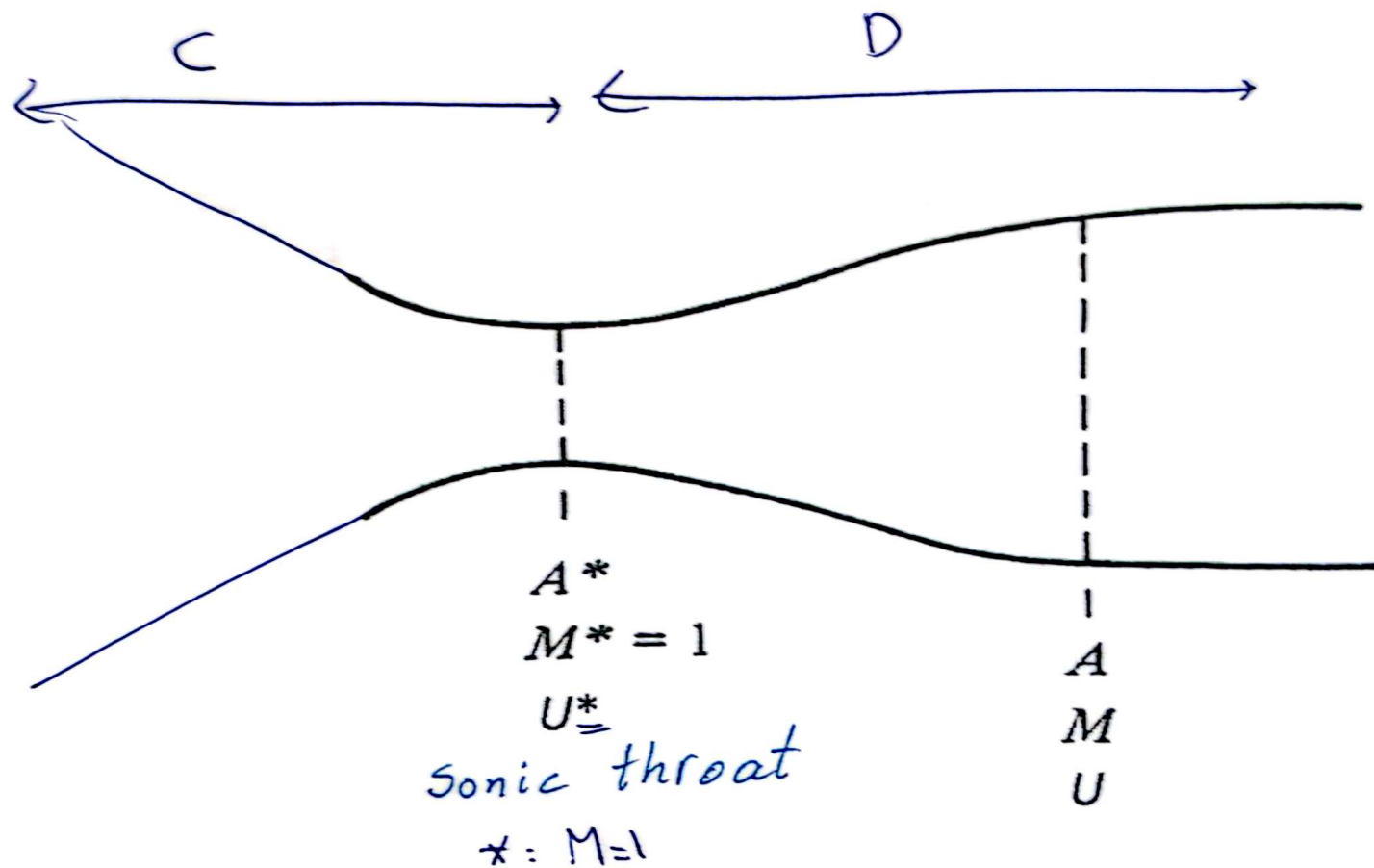
Supersonic
nozzle

only way to achieve
supersonic flow
(e.g. rockets)



Supersonic
diffuser

only way to decelerate
supersonic flow to
subsonic



* denotes sonic conditions

continuity: $\rho^* U^* A^* = \rho U A \Rightarrow \frac{A^*}{A} = \frac{\rho U}{\rho^* U^*} = \frac{p_t (p/p_t) \cdot \sqrt{\gamma R T_t} \sqrt{T/T_t} M}{p_t (p^*/p_t) \sqrt{\gamma R T^*} \sqrt{T^*/T_t}}$

use isentropic relations for * ($M=1$)

For any $\frac{A^*}{A}$, you have M at A from this Eq. 8

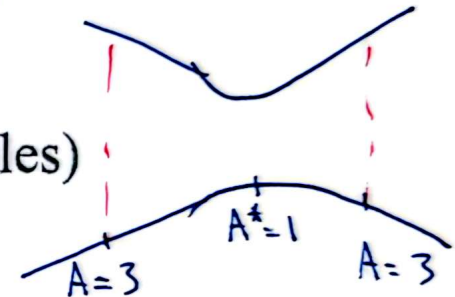
$$\Rightarrow \left(\frac{A^*}{A} \right)^2 = \frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{-(\gamma+1)/(\gamma-1)} \quad (8.43)$$

Area - Mach number relation (part of isentropic tables)

inverse: $M = f(A^* / A)$

given in isentropic tables e.g.

$$\begin{cases} \frac{A}{A^*} = 3 \\ M_{sub} = ? \end{cases} \quad \frac{A^*}{A} = 0.5 \Rightarrow \begin{cases} M = 0.31 \\ M = 2.2 \end{cases}$$



$$\frac{A}{A^*} = 3 \rightarrow M \text{ at } A = ?$$

$A^*/A < 1 \Rightarrow 2$ solutions for M

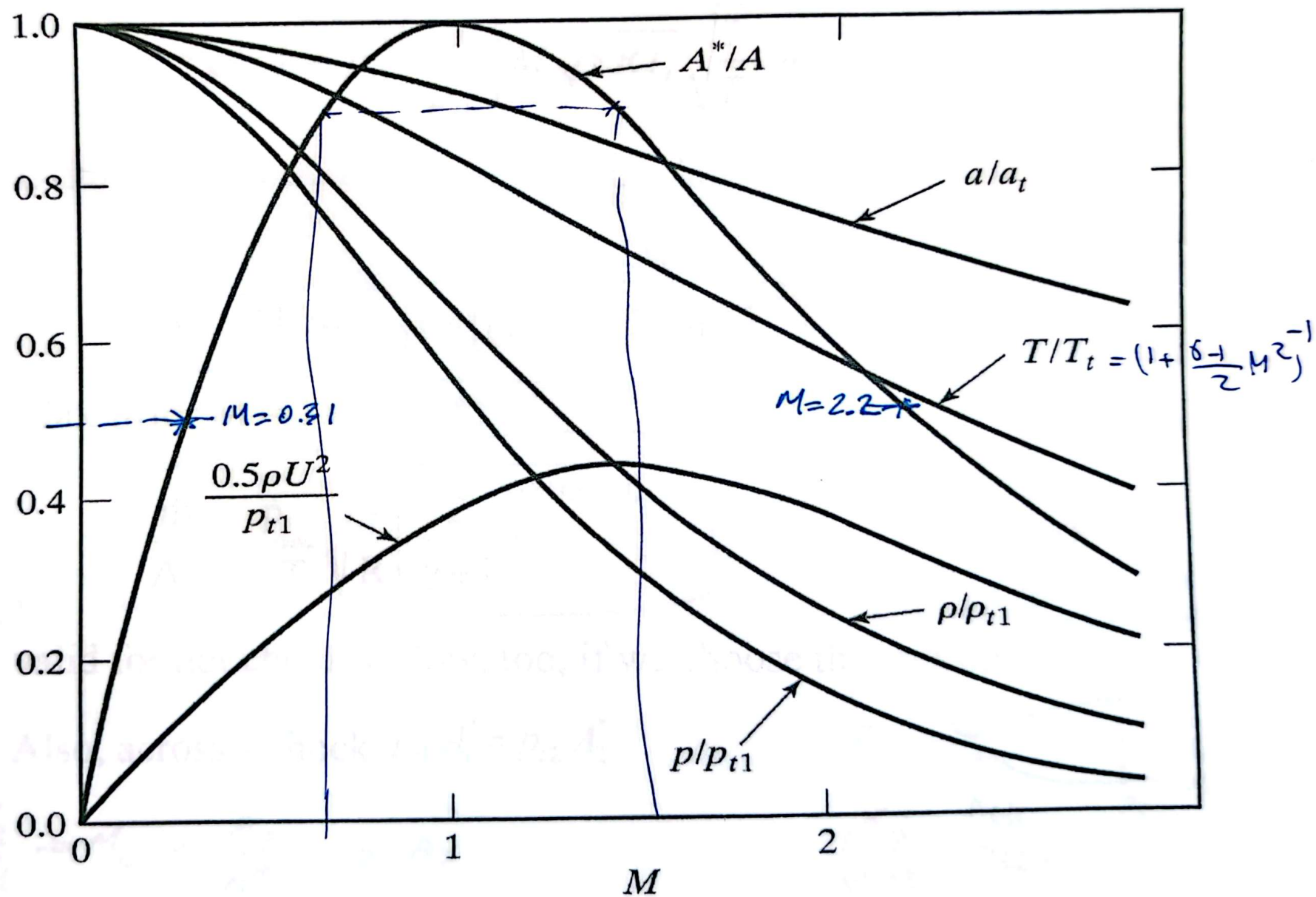
1) $M < 1$ (subsonic)

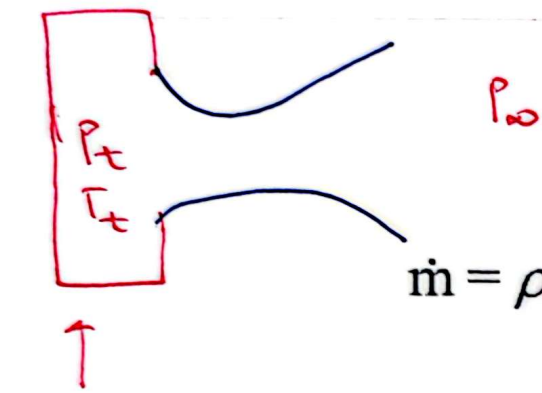
2) $M > 1$ (supersonic)

^{sup}
depends
on back
pressure

For $M < 1$ as $M \uparrow \Rightarrow A/A^*$ decreases $\downarrow$ till $M = 1$

For $M > 1$ as $M \uparrow \Rightarrow A/A^*$ increases $\uparrow$ (see graph)





reservoir

use (1) in case there is a shock ($p_{t2} < p_{t1}$)

$$\dot{m} = \rho U A = \rho_{t1} \left(\frac{\rho}{\rho_{t1}} \right) M \sqrt{\gamma R T_{t1}} \sqrt{\frac{T}{T_{t1}}} A$$

Using (8.34) and (8.37) and ideal gas law:

$$\frac{\dot{m}}{A} = \sqrt{\frac{\gamma}{R}} \frac{p_{t1}}{\sqrt{T_{t1}}} \frac{M}{\left\{ 1 + \left[(\gamma - 1) / 2 \right] M^2 \right\}^{(\gamma+1)/2(\gamma-1)}} \quad (8.44)$$

Max $\dot{m}$ for $M = 1$ ($A = A^*$)

$$\frac{\dot{m}}{A^*} = \frac{p_{t1}}{\sqrt{T_{t1}}} \sqrt{\frac{\gamma}{R} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}}$$

(8.45)

(8.45)

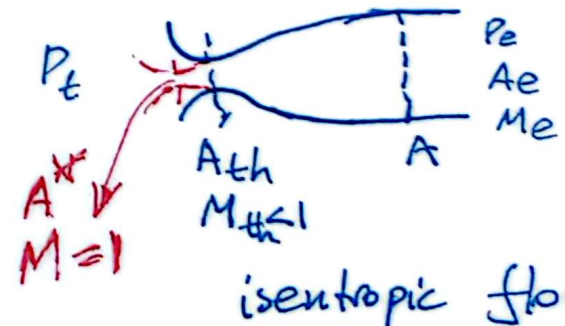
("equivalent" throat where $M_{th} = 1$)

valid for not choked flow, too, if we choose the appropriate $A^* \neq A_{th}$

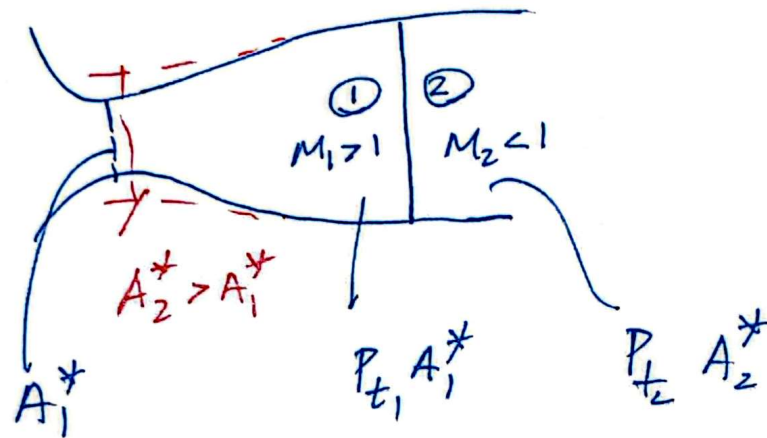
Also, across a shock $p_{t1} A_1^* = p_{t2} A_2^*$

$$\frac{p_e}{p_t} \Rightarrow M_e \sim \frac{A_e}{A^*} \sim A^*$$

$$\frac{A_{th}}{A^*} \Rightarrow M_{th} < 1$$



isentropic flow

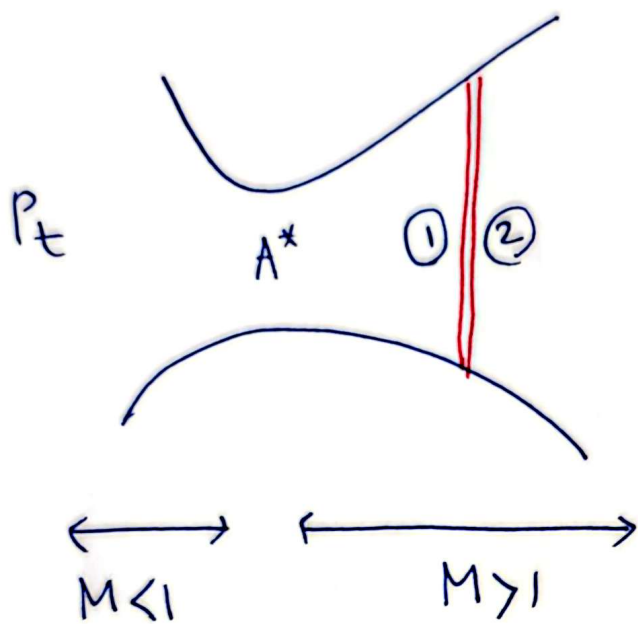


$$P_{t1} A_1^* = P_{t2} A_2^*$$

(mass flow rate is the same across the shock)

$$P_{t2} < P_{t1} \Rightarrow A_1^* < A_2^*$$

= equivalent throat area after the shock if the flow was isentropic



$$\dot{m} \propto (A^*, P_t, T_t, \gamma)$$

Across shock $\int \dot{m} = \text{cte}$
 shock is adiabatic $\rightarrow \begin{cases} T_t = \text{cte} \\ \gamma = \text{cte} \end{cases}$

So from 8.45 $\dot{m}_1 = \dot{m}_2$ results in

$$P_{t1} A_1^* = P_{t2} A_2^*$$

So when shock occurs, the flow hypothetically belongs to another nozzle with A_2^* that is greater than A_1^*

$$\text{As } P_{t2} < P_{t1}$$